

12. More on Rademacher Complexity/Covering Numbers

Friday, May 15, 2026 10:40 AM

Recall $\hat{R}_S(F) = \mathbb{E}_\sigma \sup_{f \in F} \frac{1}{m} \sum_{i=1}^m \sigma_i f(z_i)$

or, to generalize (simplify): Rademacher Complexity of a set:

def $\hat{R}(A) = \frac{1}{m} \mathbb{E}_\sigma \sup_{a \in A} \langle a, \sigma \rangle, \quad A \subseteq \mathbb{R}^m$

(By setting $A = \left\{ \begin{bmatrix} f(z_1) \\ \vdots \\ f(z_m) \end{bmatrix} : f \in F \right\}$)
 we recover $\hat{R}(A) = \hat{R}_S(F)$
 i.e. $\hat{R}_S(F) := \hat{R}(F \circ S)$

N.B. Very related ideas have been discovered/proposed in other subfields

eg. Gaussian width of a set A is $G(A) = \mathbb{E}_{g \sim N(0, I_m)} \sup_{a \in A} \langle g, a \rangle$

Also "Local Rademacher Complexity"

and Statistical Dimension $= \mathbb{E}_{g \sim N(0, I_m)} \| \text{Proj}_A(g) \|^2$

(vs. Kolmogorov complexity, length of shortest computer program...)

Calculus / tools

Lemma 26.6 [SS] $\hat{R}(c \cdot A + a_0) \leq |c| \cdot \hat{R}(A)$
 "shift-invariant", scales as expected

See Lemma 26.9 also, next page

Lemma 26.7 [SS] $\hat{R}(\text{conv}(A)) = \hat{R}(A)$

where $\text{conv}(A)$ = convex hull of A
 = smallest convex set containing A
 = intersection of all convex sets containing A
 = $\left\{ b = \sum_{i=1}^p \alpha_i \cdot a_i : \text{s.t. } p \in \mathbb{N}, a_i \in A, \alpha_i \geq 0, \sum \alpha_i \leq 1 \right\}$

Lemma 26.8 [SS] Massart Lemma Pascal Massart (b. 1958)
"t" is silent

If $|A| = N < \infty$ then $\hat{R}(A) \leq \max_{a \in A} \|a - \bar{a}\| \cdot \frac{1}{m} \sqrt{2 \cdot \log(N)}$
 $\bar{a} := \frac{1}{N} \sum_{i=1}^N a_i$ is center of mass (wlog can shift to 0)

eg. if $A = \mathcal{L} \circ \mathcal{H} \circ S$, and $\mathcal{L}$ is bounded, then $\max \|\dots\|$ is bounded

and $\hat{R}_S(\mathcal{L} \circ \mathcal{H}) = O(\gamma_m \sqrt{\log(|\mathcal{H}|)})$. Combine w/ them to bound L_D

(p. 2) More on Rademacher Complexity

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Lemma 26.9 [SS] If $\phi: \mathbb{R}^m \rightarrow \mathbb{R}^n$ is ρ -Lipschitz in each component $i \in [m]$ then $\hat{R}(\phi \circ A) \leq \rho \cdot \hat{R}(A)$

Using these tools you can sometimes compute / bound R for simple/nice enough sets

Ex $\mathcal{H}_2 := \{x \mapsto \langle w, x \rangle : \|w\|_2 \leq 1\}$, an ∞ set of linear classifiers

By Lemma 26.10 [SS], if $S_x = \{x_1, \dots, x_m\} \subseteq \mathbb{R}^n$

$$\hat{R}(\mathcal{H}_2 \circ S_x) \leq \frac{1}{\sqrt{m}} \cdot \max_i \|x_i\|_2$$

Ex $\mathcal{H}_1 := \{x \mapsto \langle w, x \rangle : \|w\|_1 \leq 1\}$

Lemma 26.11 [SS],

$$\hat{R}(\mathcal{H}_1 \circ S_x) \leq \frac{1}{\sqrt{m}} \max_i \sqrt{2 \cdot \log(2n)} \|x_i\|_\infty$$

(we might do in detail for HW)

note $\|x\|_2 \leq \sqrt{n} \cdot \|x\|_\infty$
and $\|x\|_\infty \leq \|x\|_2$

On our HW, we prove

$$\dots \leq \frac{1}{m} \|S_x\|_F^2 = \frac{1}{m} \sum \|x_i\|_2^2$$

$$\leq \frac{1}{m} \sqrt{m \cdot \max \|x_i\|_2^2}$$

$$= \frac{1}{\sqrt{m}} \max \|x_i\|_2$$

So our HW bound was sharper

Perspective

R.C. used for uniform convergence ~2000

(often tools have been around, but only "popular" once ML discovers)

R.C. mainly for classification

- For regression, if loss is bounded (less common), R.C. can work
- otherwise need specialized measures

Alternatives: (ideas like VC-dim are inherently combinatorial and won't work)

- Pseudo-dimension (like VC-dim, needs boundedness)
- Fat-shattering dim. (still needs boundedness)

see Mohri for brief discussion

(p. 3) Covering Numbers

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Another notion of complexity:

Covering Numbers

§27 [SS], §3.5 Mohri

Def The $N_p(\epsilon, A)$ covering number of a set A is the minimum number of points needed to cover A , i.e., s.t. A is contained in the union of ϵ -balls centered at these points

$$B_\epsilon(y) = \{x: \|x-y\|_p \leq \epsilon\}$$

(If p not specified, assume $p=2$)

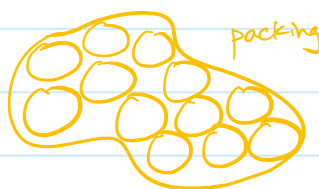
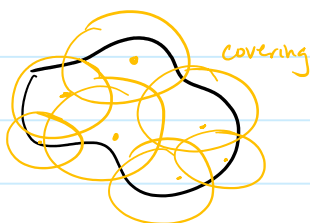
related to... packing numbers $M_p(\epsilon, A)$, the max # balls you can fit in A w/o overlapping.

of radius ϵ wrt l^p norm

$$\text{Fact: } M_p(2\epsilon, A) \leq N_p(\epsilon, A) \leq M_p(\epsilon, A)$$

In-class exercise on if

"bounded" = "totally bounded" aka precompact
Def: if $\forall \epsilon, N_p(\epsilon, A) < \infty$



We'll prove shortly that you can bound $\hat{R}(A)$ by $N_2(\epsilon, A)$

"Dudley's Thm" ('67, '87) ... and can bound N_2 by VC-dim.

uses "chaining technique" which we'll cover

Facts

$\log(\text{covering number})$ is sometimes called "metric entropy"

$$\bullet N_2(\epsilon, cA + a_0) = N_2(c\epsilon, A) \quad \text{translation/scaling}$$

$$\bullet \phi \text{ componentwise Lipschitz, then } N_2(\epsilon, \phi \circ A) \leq N_2(\epsilon/\rho, A) \quad \text{w/ constant } \rho$$

[SS] Lemma 27.1 If $A \subseteq \mathbb{R}^m$ is in a d -dimensional subspace and

$$A \subseteq B_c(0) \quad (\text{ie. radius } \leq c) \quad \text{then } N_2(\epsilon, A) \leq \left(2c \frac{\sqrt{d}}{\epsilon}\right)^d \quad \text{wrt } l^2 \text{ norm}$$

Lemma 2.2 [Woodruff] $A \subseteq \mathbb{R}^m$ is in a d -dim. subspace

and $\forall a \in A, \|a\|_2 = 1$ (ie. subspace $\cap$ sphere)
or ≤ 1 ok too

$$\text{then } N_2(\epsilon, A) \leq \left(1 + \frac{2}{\epsilon}\right)^d$$

Better than $\uparrow$
See also Tropp ACM 217 Cor 9.14 $(1 + \frac{2}{\epsilon})^d$ also
or Vershynin 9.2.13 $(\frac{2}{\epsilon})^d \leq N_2(\epsilon, A) \leq (1 + \frac{2}{\epsilon})^d$

(p. 4) More on Rademacher Complexity

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proof (fn)

wlog work in unit sphere inside $\mathbb{R}^d$.

We'll find a covering (maybe not optimal) in a greedy fashion

that will at least be maximal: nothing redundant

Let C be the centers, so covering = $\bigcup_{c \in C} B_\varepsilon(c)$

Pick $c \in A$ arbitrarily. Check if A is covered.

- If so, stop

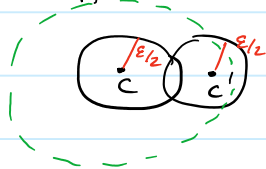
- If not, $\exists c' \in A$ that hasn't been covered
(ie. not within ε of any $c \in C$). Add c' to C

Repeat. This is maximal.

(This is the intuitive way) How big is $|C|$?

Balls of radius ε may overlap, but balls of radius $\varepsilon/2$ (around centers)

cannot overlap, otherwise non maximal.



} if $\varepsilon/2$ balls overlap
 $\Rightarrow \|c - c'\| < \varepsilon$

$\Rightarrow$ not maximal: either c (or c')
was already covered.

Furthermore, all these $\varepsilon/2$ balls are within $1 + \varepsilon/2$ of origin
(centers on unit sphere)



Recall in dimension d , volume (ball radius r) $\propto r^d$

By disjointness of $\varepsilon/2$ balls,

$$\begin{aligned} \text{Volume}(\text{all } \varepsilon/2 \text{ balls}) &= \sum_{\text{all } \varepsilon/2 \text{ balls}} \text{Vol.}(\varepsilon/2 \text{ ball}) \\ &= |C| \cdot \text{const.} \cdot (\varepsilon/2)^d \end{aligned}$$

but also

$$\text{Vol.}(\text{all } \varepsilon/2 \text{ balls}) \leq \text{Vol.}(1 + \varepsilon/2 \text{ ball}) = \text{const.} \cdot (1 + \varepsilon/2)^d$$

$$\Rightarrow |C| \cdot (\varepsilon/2)^d \leq (1 + \varepsilon/2)^d, \quad |C| \leq \left(\frac{1 + \varepsilon/2}{\varepsilon/2} \right)^d = \left(1 + \frac{2}{\varepsilon} \right)^d$$

$$\Rightarrow N_2(\varepsilon, A) \leq \left(1 + \frac{2}{\varepsilon} \right)^d. \quad \square$$

(seems loose since we didn't exploit $\|a\|=1$ only $\|a\| \leq 1$,
but in high dim. the surface is where all the volume is, so not a big deal)

Proof of lower bound,

$$(\varepsilon/2)^d \leq N_2(\varepsilon, A):$$

Cover w/ N balls, each
of radius ε . So

$$\text{Vol}(A) \leq N \cdot \text{vol}(B_\varepsilon)$$

$$\text{const.} \leq N \cdot \text{const.} \cdot \varepsilon^d$$

$$\text{so } N \geq \varepsilon^{-d} \quad \square$$